Spatial–Spectral Attention Network Guided With Change Magnitude Image for Land Cover Change Detection Using Remote Sensing Images

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***Abstract*— Land cover change detection (LCCD) using remote sensing images (RSIs) plays an important role in natural disaster evaluation, forest deformation monitoring, and wildfire destruction detection. However, bitemporal images are usually acquired at different atmospheric conditions, such as sun height and soil moisture, which usually cause pseudo and noise change in the change detection map. Changed areas on the ground also generally have various shapes and sizes, consequently making the utilization of spatial contextual information a challenging task. In this article, we design a novel neural network with a spatial–spectral attention mechanism and multiscale dilation convolution modules. This work is based on the previously demonstrated promising performance of convolutional neural network for LCCD with RSIs and attempts to capture more positive changes and further enhance the detection accuracies. The learning of the proposed neural network is guided with a change magnitude image. The performance and feasibility of the proposed network are validated with four pairs of RSIs that depict real land cover change events on the Earth's surface. Comparison of the performance of the proposed approach with that of five state-of-art methods indicates the superiority of the proposed network in terms of ten quantitative evaluation metrics and visual performance. Such as, the proposed network achieved an improvement of about 0.08%–14.87% in terms of overall accuracy (OA) for Dataset-A.**

***Index Terms*— Convolutional block attention module (CBAM), guide change magnitude image (CMI), land cover change detection (LCCD), multiscale dilation convolution module (MDCM), remote sensing images (RSIs).**

Manuscript received 9 June 2022; revised 16 July 2022; accepted 6 August 2022. Date of publication 17 August 2022; date of current version 25 August 2022. This work was supported in part by the Natural Science Basic Research Program of Shaanxi under Grant 2021JC-47; in part by the Key Research and Development Program of Shaanxi under Grant 2022GY-436; and in part by the National Science Foundation China under Grant 41971296, Grant 42122009, and Grant 61701396. *(Corresponding author: Guoqing Cui.)*

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Digital Object Identifier 10.1109/TGRS.2022.3197901

I. Introduction

REMOTE sensing images have been widely used for image classification [1], scene classification [2], target detection [3], feature extraction [4], and semantic segmentation [5]. In recent decades, land cover change detection (LCCD) referring to detecting ground change with bitemporal remote sensing images (RSIs) has become popular [6], [7]. LCCD with RSIs can estimate land cover changes on a large scale in a timely manner. Thus, LCCD has been applied successfully in natural disaster evaluation [8], land use analysis [9], and sustainable urban development [10].

Numerous LCCD methods with RSIs have been developed in the past decades [11]. Several studies have indicated that exploring spatial contextual information is important to improve detection accuracies [12], [13]. Therefore, according to the different ways of exploring spatial contextual information from RSIs, the currently available LCCD methods that are applied on RSIs can be roughly grouped into three categories: regular-region-based approaches, adaptive-neighbor-based approaches, and deep neural-network-based approaches. Each method type is reviewed below.

The most popular and intuitive approach used to utilize spatial contextual information for LCCD is the regular-region-based one with regular windows. In this approach, a regular window or strict mathematical model is adopted to explore contextual information around a pixel [12]. For example, Celik combined principal component analysis and k-means clustering with a sliding window for realizing LCCD [14]. Hong and Vatsavai proposed a sliding window-based probability change detection approach to overcome geometric and atmospheric errors from bitemporal images [15]. Saha *et al.* [16] promoted an unsupervised deep change vector analysis based on a moving window. The regular-window-based LCCD approach is intuitive and easy to be understood. Another regular-region-based approach refers to a strict mathematical model to describe contextual information around a pixel, such as multiresolution level set [17], level set evolution with local uncertainty constraint [18], Gabor-wavelet-based different measurement [19], and Markov random field [9]. Apart from a regular window, a mathematical model also describes the contextual information around a pixel in a strict manner; such a model is usually built on an assumption. Thus, a strict model usually requires some empirical parameters to produce the change detection map,

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but acquiring optical parameters for a specific dataset is time-consuming. Many studies have indicated that sliding a regular window and a strict model on an entire image for considering spatial contextual information is an effective way of smoothing noise and improving change detection accuracies [20], [21], [22]. However, this method has a disadvantage in covering the ground targets with various shapes and sizes [23].

An adaptive or multiscale-based way to explore contextual information around a pixel is expected to cover more ground targets with various shapes and sizes while exploring spatial contextual information for LCCD with RSIs. Examples of such an approach are multiscale window feature fusion for LCCD [24], [25], [26], adaptive region-based change detection approaches [27], [28], [29], and multiscale object-oriented change detection approaches [30], [31], [32]. This method has advantages in preserving the shape and boundary of the changing area because an adaptive region is composed of a group of pixels that are spatial continuously and spectral similar. The method is also suitable for covering ground targets with various shapes and sizes and more accurate than using regular window. However, selecting a suitable scale or parameter for a given dataset requires trial-and-error experiments, which are time-consuming [12], [33].

In recent years, deep learning has been demonstrated to be an attractive way to explore deep features for enhancing the performance of LCCD with RSIs [34]. One classical neural network that has been applied successfully for LCCD is named U-net [35], which allows the network to use global and contextual information simultaneously and provides better performance for LCCD [36]. Some mechanics based on the backbone of U-net are used to enhance the performance of LCCD with RSIs, such as the convolutional block attention module (CBAM) for obtaining an intermediate feature map that guides the network to learn where and what changes occur in an image scene [37], [38], [39]. The fully convolutional neural network (FCNN) has also been applied widely in LCCD with RSIs. For example, Daudt *et al.* [40] presented two FCNN architectures named fully convolutional Siamese concatenation (FC-Siam-Conc) and fully convolutional Siamese difference (FC-Siam-Diff) for LCCD with RSIs. Liu *et al.* [41] gave a novel version of the fully convolutional network (FCN) based on the local restricted convolutional layer for change detection with synthetic aperture radar (SAR) images. Zhang and Shi constructed a deep feature difference FCNN based on Visual Geometry Group (VGG) 16 for LCCD [42]. Shi *et al.* [37] developed a deeply supervised attention metric-based network (DSAMNet) for LCCD. A multiscale learning strategy is also usually suggested to explore features from multiple convolutional layers and mine effective contexture feature from an image science. For example, Lei *et al.* [43] proposed coupling FCN with pyramid pooling for landslide inventory mapping (FCN-PP). Li *et al.* [44] presented a multiscale fully convolutional network (MFCN) for LCCD. Wang *et al.* [45] proposed a deep Siamese network with multiscale feature convolution feature extraction module for change detection with multisensor RSIs. Zhang *et al.* [46] developed a neural network for building change detection with multimodality images. Liu *et al.* [47] extracted building change through local–global pyramid network and cross-task transfer learning strategy. Although numerous spatial context-based LCCD approaches have been promoted, these methods are insufficiently universal and most of them are impractical [13], [48], [49]. Therefore, the performance of LCCD with RSIs can still be improved.

The above-mentioned works from the literature indicate that detecting land cover changes with RSIs is a complicated process and the results can be enhanced by many methods. Most existing methods aim to improve the performance of LCCD tasks, and they can be summarized as follows.

1) Reducing pseudo changes from the phenological difference in bitemporal images [50], [51]: Bitemporal RSIs are acquired at different dates and seasons, seasonal phenology usually causes pseudo changes in a detection map, and conquering the pseudo change from phenological difference in bitemporal images is challenging.

2) Smoothing inherent noise by some uncertain factors [6]: Remote sensing imaging is affected by various uncertain factors, such as the effect of sensors, illumination, and atmospheric condition. These external conditions will cause noise while measuring the change magnitude between the bitemporal images and may produce many false and missed alarms in the detection map. Although denoising operation can be implemented in a preprocessing step, errors and noise are inevitable [52].

3) Covering various shapes and sizes of changed area on the ground in an image scene [12]: Intuitively, the changed area may have various shapes and sizes, and several studies have indicated that covering the shape and size accurately can improve the performance of LCCD with RSIs.

In this article, our motivation lies in enhancing the performance of LCCD with RSIs by considering the aforementioned items in a neural network. The major contributions of this article can be summarized as follows.

1) A novel neural network is designed for LCCD with RSIs. In particular, we demonstrate that the proposed novel network is feasible and robust for RSIs of low and high spatial resolution in the community of LCCD. The obtained test and comparison results indirectly demonstrate that the proposed approach is robust and feasible for different kinds of LCCD with different RSIs.

2) A change magnitude image (CMI) is proposed to be regarded as a guide image for instructing the neural network. In particular, the suggested CMI is generated by sliding an adaptive region around each pixel, which has an advantage in preserving the shape and size of a changed area. Thus, guiding the proposed neural network with the suggested CMI may inherit the advantage from CMI and enhance the detection accuracies in the learning process.

3) A CBAM and multiscale dilation convolution module (MDCM) are combined in the backbone of U-net to provide more discriminative features for recognizing change and unchanged areas with the guidance of CMI.

This article has four sections, including the introduction section. Section II describes the details of our proposed

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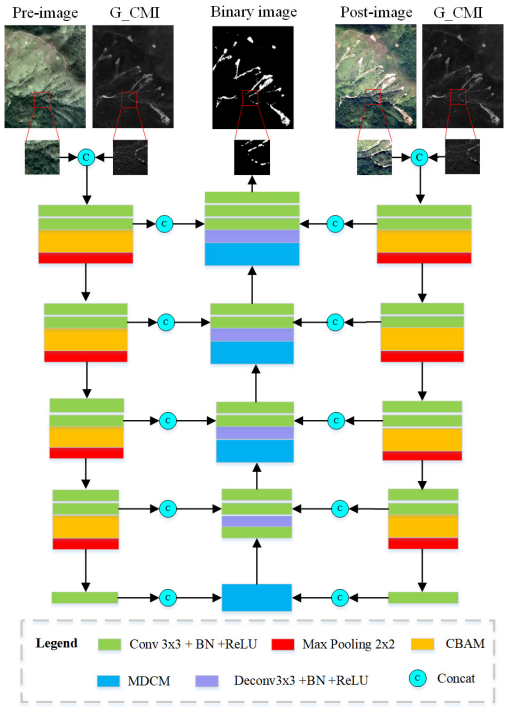


Fig. 1. Framework overview of the proposed approach.

approach. Experiments are conducted and presented in Section III. A conclusion is given in Section IV.

II. Methodology

A. Framework Overview

Fig. 1 shows that the proposed approach has a similar structure to FC-Siam-Conc [40] and FC-Siam-Diff [40], and both of them include two encoder modules that are symmetric to the decoder. The difference between the two algorithms is that the basic unit of FC-Siam-Conc [40] and FC-Siam-Diff [40] is the convolution operation, while the basic unit of the proposed approach is the convolution operation coupled with the enhanced CBAM and MDCM. Moreover, the parallel encoding process is guided by an additive CMI.

The proposed approach contains the encoder, fusion, and decoder. The inputs for the proposed approach contain two parts, bitemporal images and the CMI. The output is the binary change detection map. Each layer of the encoder contains two 3×3 convolutions, a 2×2 max pooling, a CBAM, and a MDCM to learn and generate high discriminative features for detecting change. In the fusion process, we mainly use MDCM to fuse the bitemporal features and expose the changed and unchanged areas with the guidance of CMI. The decoder leverages linear projection to reduce the feature dimension to 2, and this operation generates a binary change detection map. Details of each major process will be presented in detail below.

B. Generation of Guide CMI

A CMI describes the change distance between bitemporal images, and the CMI is suggested to be a guide image to enhance the learning process in the proposed approach. Using

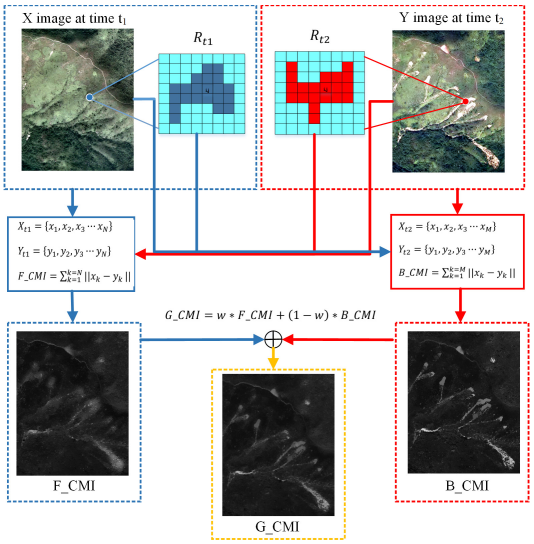


Fig. 2. Generation flowchart of guide CMI (G_CMI).

contextual information is beneficial in preserving the shape and smoothing the noise of a ground target [27], [53]. Thus, a novel CMI generation method based on the sliding adaptive region is developed here. The flowchart of this approach is illustrated in Fig. 2, and it can be split into three parts as presented below.

1) Constructing an adaptive region around each pixel of the bitemporal image: In addition to various approaches [27], [29], [54], adaptively using spatial contextual information is beneficial for improving the performance of LCCD with RSIs. Here, an automatic adaptive region extension algorithm that has been applied for detecting landslides is used to generate an adaptive region around each pixel of bitemporal images [55]. In the adopted approach, if $p_1(i, j)$ is a pixel of the preevent image(X), and the adaptive region around $p_1(i, j)$ is extended iteratively according to two extensions rules, $p_{\text{sur}} \in [m - \text{std}, m + \text{std}]$ and $|p_{\text{sur}} - p_1(i, j)| < \text{std}$, where m and std indicate the mean and standard deviation of the pixels within an extended region, respectively, and p_{sur} is the neighboring pixels around $p_1(i, j)$. In the extension, m and std are updated iteratively. An extension around a pixel is terminated until one of the conditions is unsatisfied, and an adaptive region around a pixel is generated.

2) Calculating the forward CMI (F_CMI) and backward CMI (B_CMI) based on the corresponding adaptive region. Given the difference between the bitemporal images in terms of atmospheric conditions and season phenology, the adaptive regions around the pixel (i, j) of the bitemporal images are generally different in shape and size. We define F_CMI and B_CMI for symbolizing the forward CMI and backward CMI to solve the inconsistency. F_CMI indicates the change distance between the bitemporal images and is calculated with the adaptive region generated from preevent images. Thus, $F_\text{CMI} = \sum_{k=1}^{k=N} \|x_k - y_k\|$, where $X_{t_1} = \{x_1, x_2, x_3, \ldots, x_N\}$ and $X_{t_2} = \{x_1, x_2, x_3, \ldots, x_N\}$ are the pixels of the bitemporal images. These pixels are within the adaptive region that is generated based on the preevent images. N denotes the total

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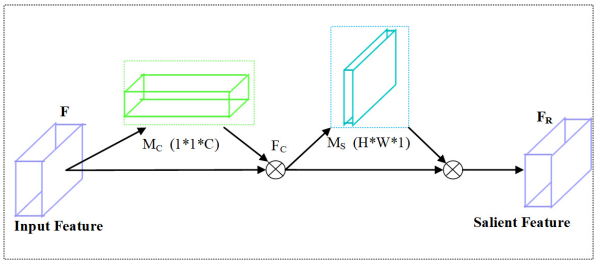


Fig. 3. Overview of the CBAM block.

pixels within the adaptive region. The adaptive region is obtained from postevent images, and B_CMI can be obtained in the same rule.

3) Fusing F_CMI and B_CMI into a guide image. To emphasize the changed area, F_CMI and B_CMI are fused into a guide image as follows: $G_\mathrm{CMI} = \omega \times F_\mathrm{CMI} + (1 - \omega) \times B_\mathrm{CMI}$, where G_CMI is the guide CMI, ω is a weight to balance the effect from F_CMI and B_CMI for G_CMI, and $\omega = (\mathrm{com}(R_{t_1}, R_{t_2})/N + M)$, where N and M denote the total pixels within the adaptive regions R_{t_1} and R_{t_1}, respectively. $\mathrm{com}(R_{t_1}, R_{t_2})$ indicates the number of equal pixel pairs within the adaptive regions R_{t_1} and R_{t_1} around a position (i, j).

C. Convolutional Block Attention Module

On the basis of the definition of guide CMI, we introduce the CBAM to learn more distinguishable features to divide the image into change and unchanged areas. As shown in Fig. 3, the CBAM contains two submodules, namely, a channel attention submodule (M_c) and a spatial attention submodule (M_s), to help emphasize the learning of changed and unchanged information in the extracted features in different dimensions. The channel attention submodule pays more attention to capturing a long range of contextual information among channels. If the input feature map is $F \in R^{C\times H\times W}$, then the process of a channel attention submodule can be formularized using (1)

$$M_c(F) = \sigma\left(\mathrm{MLP}^r(\mathrm{AvgPool}(F)) + \mathrm{MLP}^r(\mathrm{MaxPool}(F))\right) \tag{1}$$

where F symbolizes an input feature of size $C \times H \times W$. An average pooling layer is applied to construct two aggregated vectors with the size $C \times 1 \times 1$. A lightweight sharing multilayer perception (MLP) module with a channel reduction ratio r is applied to each vector to acquire the weight of each channel. The MLP module is composed of two 1×1 convolutional layers: the first layer reduces the input channels times, and the second layer restores the reduced channel to the original input. The two layers are hooked by a rectified linear unit (ReLU) layer. The intermediate channel feature can be obtained by elementwise sum with a sigmoid function σ. The revised feature F_c is generated by the original feature multiplied with the channel refined feature

$$F_c = M_c(F) \otimes F, F \in R^{C\times H\times W}, M_c \in R^{C\times 1\times 1}. \tag{2}$$

In the spatial attention submodule, as shown in (3), a refined channel feature F_c is similarly processed with average pooling and max pooling to concentrate on mining shape and structure feature of the change detection area. Then, two $1 \times H \times W$ matrixes are concatenated and forwarded into a convolution layer with a 3×3 kernel size. Finally, a sigmoid function is used to obtain the final refined spatial attention feature map $M_s(F_c)$

$$M_s(F_c) = \sigma\left(f^{3\times 3}\left(\left[\mathrm{AvgPool}(F_c), \mathrm{MaxPool}(F_c)\right]\right)\right). \tag{3}$$

Based on the results of the channel and spatial attention submodules, the final revised feature (F_R) can be obtained using (4)

$$F_R = M_s(F_c) \otimes F_c, M_s \in R^{1\times H\times W}, F_c \in R^{C\times H\times W}. \tag{4}$$

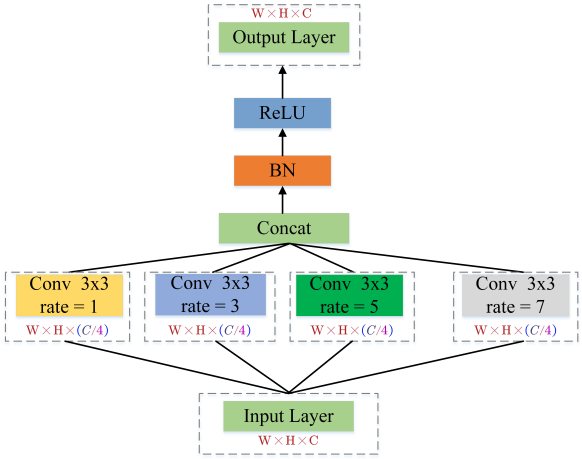


Fig. 4. Overview of MDCM block.

D. Multiscale Dilation Convolution Module

In the encoding process, deep discriminable feature of bitemporal images is explored with CBAM that is guided by F_CMI and B_CMI. However, in the real world, changed and unchanged areas may have different shapes and sizes. Therefore, a simple but effective module called MDCM is used in the decoding process to enlarge the receptive field and capture multiscale changed and unchanged areas. As shown in Fig. 4, the input features are filtered with different convolutional dilatated rations. Then, the receptive field of the convolutional layers can be revised into different sizes. The receptive field size is calculated by receptive field size (RFS) $= (k - 1) \times (r - 1) + k$, where k is the convolutional filter size and is the dilatated rate. Finally, convolutional layers with different receptive field sizes are concentrated together to obtain a larger receptive field. The suggested MDCM is embedded in each layer of the decoding process, as presented in Fig. 4.

Based on the above-mentioned neural network design and due to that the change detection can be regarded as a binary classification problem, a binary cross-entropy is selected as the loss function for our proposed neural network. If p is the prediction probability of change, and the prediction probability of unchanged is $(1-p)$, then the proposed binary cross-entropy loss function can be marked as

$$\mathrm{Loss} = \frac{1}{N}\sum_i -\left[y_i \log p_i + (1 - y_i)\log(1 - p_i)\right] \tag{5}$$

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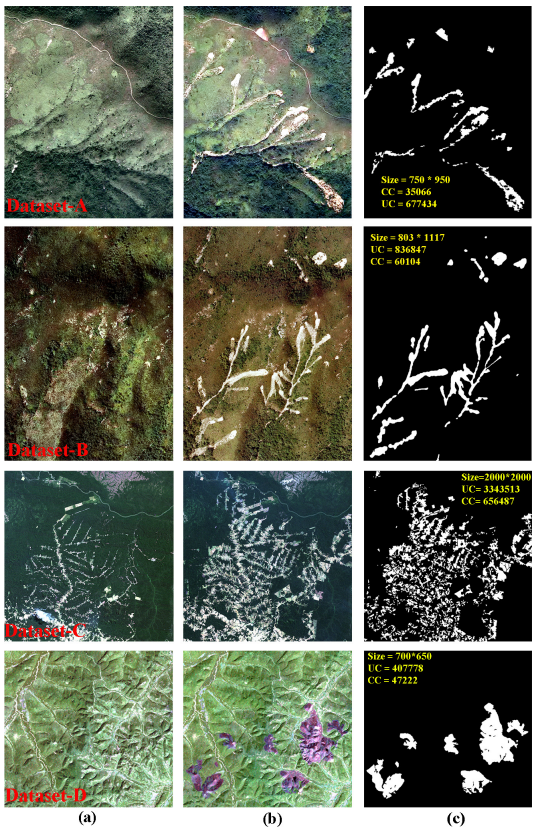


Fig. 5. Datasets A-D. (a) Preevent image. (b) Postevent image. (c) Ground reference map.

where y_i is the label of the ith sample, and $y_i = 0$ and $y_i = 1$ are, respectively, the unchanged and changed samples, and p_i is the prediction probability for change class.

III. Experiments

In this section, the performance and feasibility of the proposed approach are verified by comparing the approach with some cognate state-of-art methods by some real land cover change events. We first briefly describe the referred datasets. Then, five state-of-art methods are introduced for comparison. Finally, the experimental results are analyzed.

A. Data Description

Four pairs of datasets include aerial photographs with very high spatial resolution and Landsat satellite images with a 30 m/pixel resolution, as shown in Fig. 5. Details of each dataset are presented below:

The first and second datasets are denoted as Dataset-A and -B in the experiments, respectively. These datasets depict the landslide sites in Lantau Island in the Hong Kong area, China. The bitemporal images of Dataset-A were acquired in April 2007 and July 2014, and the bitemporal images of Dataset-B are obtained in 2006 and 2011 (the month cannot be noted in the metadata). They both have a spatial resolution of 0.5 m/pixel. The challenge of these datasets is that they have a very high spatial resolution, which leads to salt-and-pepper noise in the detection map. Moreover, the studying areas are forests, and external imaging conditions, such as sun height, may cause noise in the detection map. Some gravels are also confused with the landslide sites in the studying area.

The third dataset is referred as Dataset-C. This dataset was acquired by Landsat-5 in July 2000 and July 2006, the size of the image scene is 2000 × 2000 with a 30 m/pixel resolution, and it depicts forest deformation in the Amazon forest in Rondónia, Brazil. As shown in Fig. 5, bands (3-2-1) were used for composing false-color image and conducting experiments. The image scene covers a forest deformation in Rondónia, Brazil, of the Amazon area. The bitemporal images were acquired on July 15, 2000, and July 16, 2006.

The fourth dataset is marked as Dataset-D. This dataset was acquired by the Landsat-5 Thematic Mapper with a 30 m/pixel resolution, and bands (3-2-1) are used for detecting change. This dataset captures forest deformation that is caused by wildfire at the border of NeiMeng and Heilongjiang provinces, China. The postevent image was obtained on June 2001 and 2017, and the preevent image was acquired on August 22, 2006.

B. Experimental Setting

Five state-of-the-art LCCD methods are selected here for comparison to verify the superiority of the proposed approach. The characteristics of each approach are briefly introduced below.

1) Classical U-net [35]: The classical U-net is based on the encoding and decoding symmetric branches. The U-net is trained end to end from bitemporal images and has been shown to outperform a binary prediction map.

2) FC-Siam-Conc [40]: This neural network is an extended version of the fully convolutional early fusion (FC-EF) net, which adapts the U-net model into a full convolutional operation. In contrast to the traditional U-net, FC-EF has only four max pooling and four upsampling layers, instead of five layers in the U-net backbone. The difference between FC-Siam-Conc, as an extension of FC-EF, and FC-EF lies in the way to achieve the skip connection. Specifically, FC-Siam-Conc skips two connections during the decoding steps, and the extracted feature map is processed one by one in the encoding stream.

3) FC-Siam-Diff [40]: In the above-mentioned FC-Siam-Conc [40], change detection concentrates on distinguishing the difference between the bitemporal images. Instead of concatenating connections from the encoding branches, FC-Siam-Diff calculates the magnitude image by image difference in the learning process [40].

4) Fully convolutional network with pyramid pooling (FCN-PP) [43]: One motivation of FCN-PP is to cover various ground targets with multiscale shapes and sizes. To achieve this motivation, FCN-PP adopts a pyramid pooling into the backbone of U-net, and it has an advantage in obtaining landslide inventory mapping with high accuracies.

5) Pseudo Siamese convolutional neural network (PSI-CNN) [46]: This neural network is developed for detecting building change detection with multimodal RSIs. It has three convolutional blocks and three fully connected layers, and each convolutional block is followed by an ReLU as activation function. A max-pooling layer is also added to each convolutional

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TABLE I

COMPARISON BETWEEN THE PROPOSED APPROACH AND THE FIVE STATE-OF-ART METHODS FOR DATASET-A AND -B, KA∈ [0, 1], AND OTHERS ARE PRESENTED IN PERCENTAGE (%)

Methods		OA	AA	Ka	FA	MA	TE	UCIoU	CIoU	mIoU	F1-Score
U-net [35]	Dataset-A	98.03	85.19	0.77	**0.57**	29.05	1.97	63.96	97.96	80.96	78.02
	Dataset-B	98.68	90.28	0.88	**0.02**	19.43	1.32	80.37	98.61	89.49	89.12
Siam-U-net-Conc [40]	Dataset-A	97.96	92.52	0.80	1.45	13.52	2.04	67.56	97.87	82.71	80.64
	Dataset-B	98.58	90.66	0.88	0.20	18.47	1.42	79.33	98.49	91.00	88.47
Siam-U-net-Diff [40]	Dataset-A	98.06	88.01	0.79	0.84	23.15	1.94	66.14	97.99	82.07	79.62
	Dataset-B	97.39	84.25	0.77	0.58	30.92	2.61	63.92	97.26	80.59	77.99
PSI-CNN [46]	Dataset-A	97.87	86.4	0.77	0.88	26.32	2.13	62.96	97.79	80.37	77.27
	Dataset-B	98.30	89.94	0.86	0.40	19.71	1.70	76.02	98.21	87.11	86.37
FCN-PP [43]	Dataset-A	83.36	83.49	0.27	16.66	16.37	16.6	19.83	82.64	51.24	33.09
	Dataset-B	96.98	89.33	0.77	1.84	19.50	3.02	64.09	96.81	80.45	78.12
MFCN [44]	Dataset-A	98.16	89.98	0.80	0.95	19.09	1.84	68.38	98.08	83.23	81.22
	Dataset-B	98.66	93.37	0.89	0.52	12.74	1.339	81.37	98.58	89.97	89.73
DSAMNet [37]	Dataset-A	98.15	87.30	0.79	0.66	24.73	1.85	66.72	98.08	82.40	80.04
	Dataset-B	98.28	94.45	0.77	1.13	9.96	1.72	77.80	98.17	87.99	87.52
Proposed method	Dataset-A	**98.23**	**93.21**	**0.82**	1.23	**12.36**	**1.77**	**70.87**	**98.15**	**84.51**	**82.95**
	Dataset-B	**98.74**	**97.92**	**0.90**	1.14	**3.023**	**1.26**	**83.73**	**98.65**	**91.19**	**91.14**

layer to add translation invariance to the model and decrease the number of parameters to learn.

6) MFCN [44]: This network concentrated on solving the imbalance of positive and negative samples' problem based on multiscale fully convolutional neural network (MFCN). Moreover, a loss function combining weighted binary cross-entropy loss and dice coefficient loss is also used in MFCN. Its performance has been verified with real RSIs.

7) DSAMNet [37]: This network was developed to reduce the effect from illumination, noise, and scale, which usually caused pseudo-changes and noise in a detection map. To achieve the motivations, a deeply supervised (DS) attention metric-based network is developed based on metric learning and CBAMs to explore more discriminative features for LCCD.

In summary, all the methods mentioned above belong to the group of U-net networks. To guarantee comparative fairness, the ratio of the training and test data for each dataset and method used in comparison is 80% and 20%, respectively, and the learning rate is set to 0.0001. The Adam optimizer [56] is also adopted to accelerate the learning process. Then, each method is applied to the same dataset for comparing the performance and effectiveness. In addition, all the experiments were implemented with a PyTorch as the backend, which is powered by a workstation with an Intel(R) Xeon (TM) i7-2123 central processing unit (CPU) at 3.6 GHz, 32 GB of RAM, and a single NVIDIA GeForce RTX 2080 Ti graphics processing unit (GPU).

Ten widely used metrics, including false alarm (FA), missing alarm (MA), total error (TE), overall accuracy (OA), Kappa coefficient (Ka), average accuracy (AA), complete IoU (CIoU), user complete intersection over union (UCIoU), mIoU, and F1-score, are adopted in the experiments to quantitatively evaluate the performance of the proposed approach, and the descriptions of each metric are detailed in [12].

C. Comparative Results

On the basis of the above-mentioned parameter set, the results obtained by different methods are compared as follows:

1) Dataset-A and -B: The quantitative results for the ten metrics of all the methods are summarized in Table II. As shown in the table, the proposed approach achieves best detection accuracies on nine out of ten metrics for Dataset-A and -B. For example, compared with the five state-of-art methods, the proposed approach obtains 98.23% and 98.74% for Dataset-A and -B in terms of OA., respectively When observing the proposed approach in terms of FA and compared with that of Siam-U-net-Diff [40], U-net [35], and PSI-CNN [46], the proposed approach does not obtain the highest but has a competitive accuracy of 1.23% for Dataset-A in terms of FA. A similar comparative observation can be found for Dataset-B. The visual performance comparisons for Dataset-A and -B further support the quantitatively comparative conclusion, as demonstrated in Figs 6 and 7, respectively. The proposed approach misses the least changed pixel among all the methods, as highlighted by the red color in each result.

2) Dataset-C and -D: The RSIs with very high resolution and median-low resolution are different in terms of spectral information, spectral variability, spatial displacement, and information loss level. In the experiments, the proposed approach and the five state-of-art cognate methods were applied on Dataset-C and -D to further investigate the performance and adaptability of the proposed approach. Dataset-C was acquired by Landsat-5 in July 2000 and July 2006, the size of the image scene is 2000 × 2000 pixels, and it depicts forest deformation in the Amazon forest in Randonia, Brazil; Dataset-D is a Landsat-5 TM image which depicts the wildfire on the border of NeiMeng and Heilongjiang Provinces, China. These datasets comprise low-median (30 m/pixels) satellite images that are different from the very high spatial resolution

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TABLE II

COMPARISON BETWEEN THE PROPOSED APPROACH AND THE FIVE STATE-OF-ART METHODS FOR DATASET-C AND -D, KA∈ [0, 1], AND OTHERS ARE PRESENTED IN PERCENTAGE (%)

Methods		OA	AA	Ka	FA	MA	TE	UCIoU	CIoU	mIoU	F1-Score
U-net [35]	Dataset-C	92.07	82.58	0.69	3.30	31.58	7.92	58.62	91.06	74.84	73.91
	Dataset-D	96.12	92.7	0.80	2.99	11.61	3.88	70.25	95.72	82.99	82.53
Siam-U-net-Conc [40]	Dataset-C	93.51	89.64	0.77	4.60	16.13	6.49	67.96	92.48	80.22	80.92
	Dataset-D	97.43 9	6.46	0.87	2.32	4.76	2.57	79.35	97.15	88.25	88.49
Siam-U-net-Diff [40]	Dataset-C	91.96	78.45	0.66	**1.44**	41.66	8.04	54.36	91.11	72.73	70.43
	Dataset-D	98.57	96.8	0.92	0.96	5.44	1.43	87.30	98.42	92.86	93.22
PSI-CNN [46]	Dataset-C	93.24	85.88	0.74	3.16	25.08	6.76	64.54	92.3	78.42	78.45
	Dataset-D	98.18	94.3	0.90	**0.80**	10.59	1.82	83.61	98.00	90.8	91.08
FCN-PP [43]	Dataset-C	87.22	87.31	0.62	12.82	**12.57**	12.78	52.89	85.08	68.99	69.19
	Dataset-D	91.89	68.55	0.46	2.00	60.90	8.14	33.34	91.54	62.44	50.01
MFCN [44]	Dataset-C	92.95	90.72	0.76	5.96	12.60	7.05	67.06	91.77	79.42	80.28
	Dataset-D	98.59	94.63	0.92	0.38	10.35	1.42	86.80	98.44	92.62	92.93
DSAMNet [37]	Dataset-C	93.02	86.89	0.74	3.98	22.23	6.98	64.65	92.00	78.32	78.53
	Dataset-D	98.31	97.12	0.91	1.38	4.39	1.69	85.46	98.13	91.79	92.16
Proposed method	Dataset-C	**93.58**	**90.27**	**0.77**	4.80	14.65	**6.42**	**68.57**	**92.53**	**80.55**	**81.36**
	Dataset-D	**98.61**	**98.42**	**0.93**	1.34	**1.83**	**1.39**	**87.99**	**98.45**	**93.22**	**93.61**

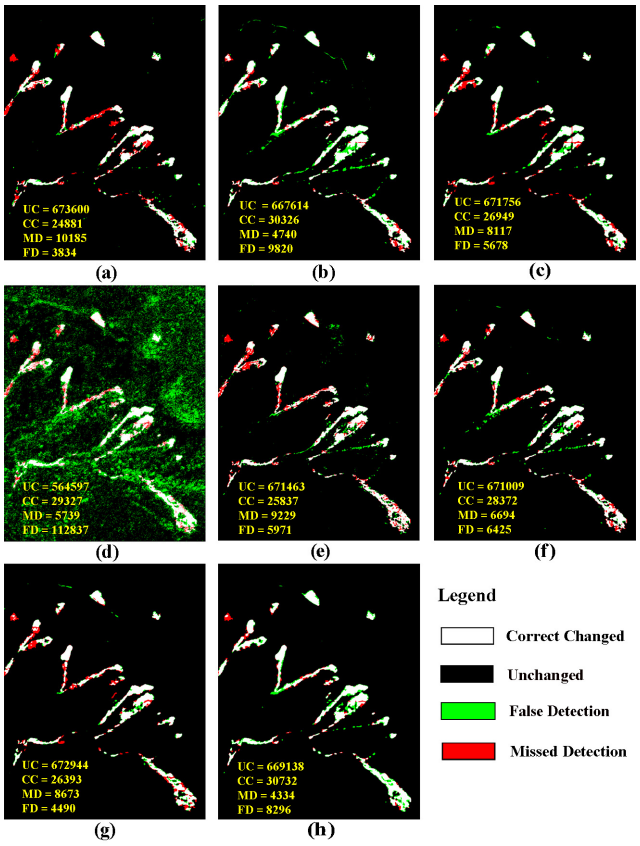


Fig. 6. Binary change detection maps obtained by different LCCD methods for Dataset-A. (a) U-net [35]. (b) Siam-U-net-Conc [40]. (c) Siam-U-net-Diff [40]. (d) FCN-PP [43]. (e) PSI-CNN [46]. (f) MFCN [44]. (g) DSAMNet [37]. (h) Proposed method.

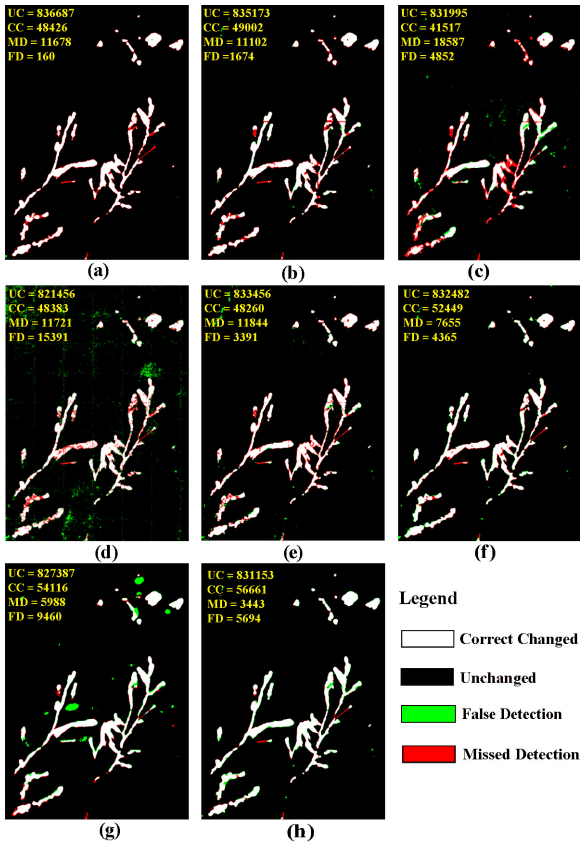


Fig. 7. Binary change detection maps obtained by different LCCD methods for Dataset-B. (a) U-net [35]. (b) Siam-U-net-Conc [40]. (c) Siam-U-net-Diff [40]. (d) FCN-PP [43]. (e) PSI-CNN [46]. (f) MFCN [44]. (g) DSAMNet [37]. (h) Proposed method.

images (Dataset-A and -B). The quantitative assessment of the results is presented in Table III for Dataset-C. As observed, the proposed approach achieves the best accuracies for eight out of ten metrics among those of other methods. For example, the proposed method obtains 93.58% (OA), 90.27% (AA), and 0.77 (Ka) for Dataset-C, and its performance is the

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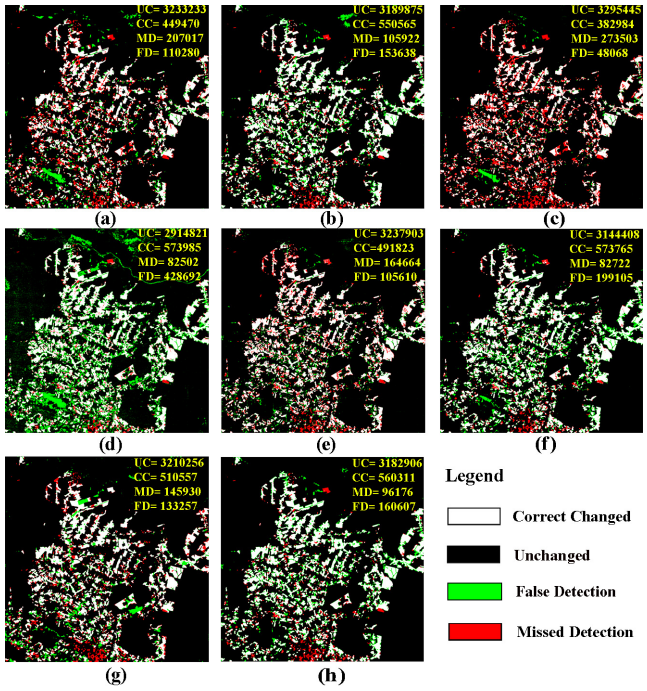


Fig. 8. Binary change detection maps obtained by different LCCD methods for Dataset-C. (a) U-net [35]. (b) Siam-U-net-Conc [40]. (c) Siam-U-net-Diff [40]. (d) FCN-PP [43]. (e) PSI-CNN [46]. (f) MFCN [44]. (g) DSAMNet [37]. (h) Proposed method.

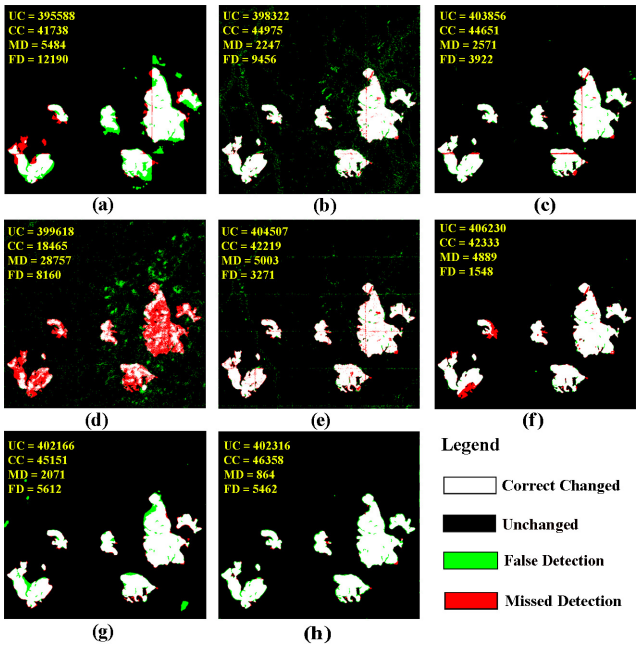


Fig. 9. Binary change detection maps obtained by different LCCD methods for Dataset-D. (a) U-net [35]. (b) Siam-U-net-Conc [40]. (c) Siam-U-net-Diff [40]. (d) FCN-PP [43]. (e) PSI-CNN [46]. (f) MFCN [44]. (g) DSAMNet [37]. (h) Proposed method.

highest compared with that of the other approaches. The visual performance comparisons shown in Fig. 8 verify the conclusion from the above-mentioned quantitative evaluation. When these approaches are applied on Dataset-D, a similar conclusion can be obtained. As shown in Table III and Fig. 9, quantitative comparisons and visual performance observations for Dataset-D demonstrate the superiority of the proposed approach in realizing the LCCD task.

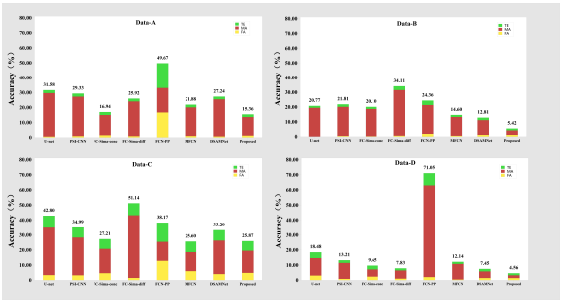


Fig. 10. Bar chart comparisons of various methods for detection accuracies using FA, MA, and TE for Dataset-A to -D.

TABLE III
EFFICIENCY COMPARISONS FOR EACH METHOD AND DATASET (IN SECONDS)

Methods	Data-A	Data-B	Data-C	Data-D
U-net [35]	312.97s	311.74s	337.60s	299.78s
Siam-U-net-Conc [40]	164.95s	183.01s	193.13s	158.05s
Siam-U-net-Diff [40]	109.08s	878.08s	112.06s	127.01s
PSI-CNN [46]	1358.24s	1446.61s	1496.07s	1306.04s
FCN-PP [43]	878.07s	924.01s	914.78s	898.28s
MFCN [44]	398.26s	142.09s	676.33s	146.63s
DSAMNet [37]	133.61s	183.12s	583.63s	181.52s
Proposed method	214.00s	258.02s	289.61s	253.01s

Some negative evaluations (FA, MA, and TE) and positive evaluations (OA, mIoU, CIoU, and F1-score) are summarized in Figs. 10 and 11 for further demonstrating the comparison between the proposed approach and the other state-of-art methods. This summary indicates the advantages of the proposed approach. Based on the above-mentioned experimental comparison with different datasets, including two pairs of aerial photographs with a very high spatial resolution (0.5 m/pixel) and two pairs of Landsat satellite images with a median-low spatial resolution (30 m/pixel), these application and result comparisons indicate that the proposed neural network has an advantage in achieving better detection accuracies for the LCCD task compared with the five state-of-art cognate methods. In addition, as shown in Fig. 10, the y-axis gives accuracy(%) for the summary of the negative evaluations (FA, MA, and TE), and the proposed approach demonstrated the lowest bar while compared with that of other methods. The visual performance comparisons verify that the proposed network is superior for smoothing noise and improving the detection accuracies.

D. Discussion

In this section, the relationship between the detection accuracies and the parameters of the proposed neural network is discussed to promote its application in practice.

First, the sensitivity between the batch size and the detection accuracies in terms of UCIoU, CIoU, F1-score, and mIoU

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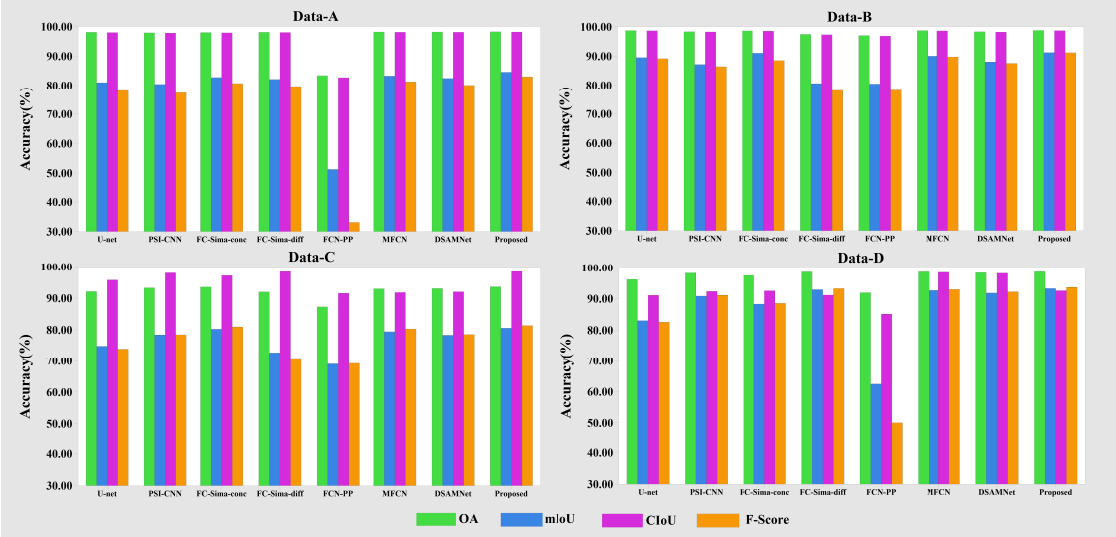


Fig. 11. Bar chart comparisons of various methods for detection accuracies using OA, mIoU, CIoU, and F1-score for image scenes of Dataset-A to -D.



Fig. 12. Relationship between batch sizes and CIoU, mIoU, F1-score, and UCIoU for Dataset-A, -B, -C, and -D for the proposed network.

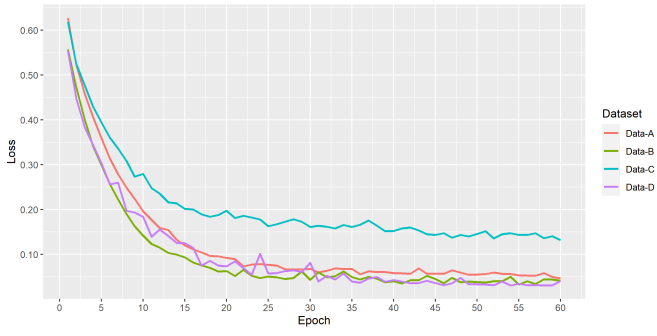


Fig. 13. Relationship between the number of training epochs and loss rate for Dataset-A, -B, -C, and -D for the proposed network.

is discussed. As shown in Fig. 12, the sensitivity between the detection accuracies and the batch size of feeding the proposed network is different for each dataset in terms of a specific metric. The proposed network also gives different accuracies when it is applied to different datasets based on the same metric, and the variation degree for a metric with an increment of batch size varies for different datasets. For example, the UCIoU of the proposed network ranges from 64.56% to 70.87% with the increment in batch size from 32 to 128 for Dataset-A. However, it ranges from 69.28% to 86.12% for Dataset-C with the same increment in batch size. UCIoU performs at 64.56%, 64.8%, 69.28%, and 88.7% when it is applied on Dataset-A, -B, -C, and -D, respectively. Therefore, when the proposed network is applied to a dataset, the optimal batch size for an evaluation metric should be obtained by trial-and-error experiments.

Second, the loss rate between the number of training epochs for each dataset of our proposed network needs to be discussed. As shown in Fig. 13, the loss rate of the proposed approach for each dataset is different at the same epoch number. However, they all converge gradually with the increase in the number of epochs. The loss rate and the convergence of the proposed approach are different for each dataset. Thus, the optimal number of epochs for the proposed network is determined by the application dataset.

Third, the efficiency of each approach was compared in this section. From the efficiency comparisons in Table III, it can be found that the proposed approach provided a competitive accuracy in terms of efficiency. In addition, the efficiency of the proposed approach is varied with the inputting dataset.

IV. Conclusion

In this article, a new deep-learning-based approach is proposed to achieve LCCD tasks with RSIs. The main contributions of the proposed approach lie in developing an approach for generating a CMI with a sliding adaptive region and constructing a neural network that is guided by the suggested CMI and spatial–spectral attention modules for enhancing the learning performance. The advantages of the proposed neural network are summarized as follows.

1) Advanced detection accuracies are obtained with the proposed approach. The comparative case studies based on four

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pairs of real RSIs indicate that the proposed approach outperforms five widely used cognate methods, namely, U-net [35], Siam-U-net-Conc [40], Siam-U-net-Diff [40], FCN-PP [43], PSI-CNN [46], MFCN [44], and DSAMNet [37], in terms of visual performance and ten quantitative evaluation metrics.

2) Coupling CMI to guide spatial–spectral attention modules in a neural network is effective in inhibiting the pseudo change noise and preserving the shape of changed areas. Owing to the guidance of CMI, the proposed neural network with the spatial–spectral attention modules can learn changed and unchanged areas with less noise. The CMI is also generated with a sliding adaptive region, which will be beneficial for preserving the shape and size of a changed area. This result may be universal for reducing the pseudo change of LCCD with RSIs while using a deep learning network.

3) A simple, robust, and no-parameter approach is preferred and acceptable for practical engineering applications. The proposed approach has no parameters and threshold for obtaining a binary change detection map. Moreover, it is suitable for detecting land cover change with very high spatial resolution or low-median spatial resolution RSI. This characteristic makes it easy to be applied in practice.

Despite the advantages of the proposed neural network summarized above, the practicality and widespread application with large areas and other kinds of datasets should be further investigated. In the future study, we will concentrate on collecting more kinds of RSIs with large size for investigating comprehensively the performance of the proposed approach. In addition, the interpretability of the learning process of the proposed approach should also be considered further in our future research.

ACKNOWLEDGMENT

The authors are grateful to the Editor-In-Chief, an Associate Editor, and the Reviewers for their insightful comments and suggestions.

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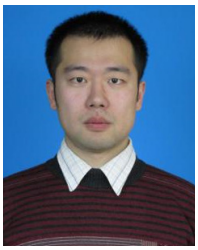
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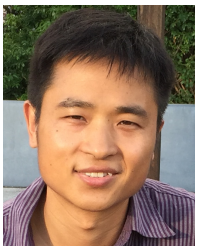
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